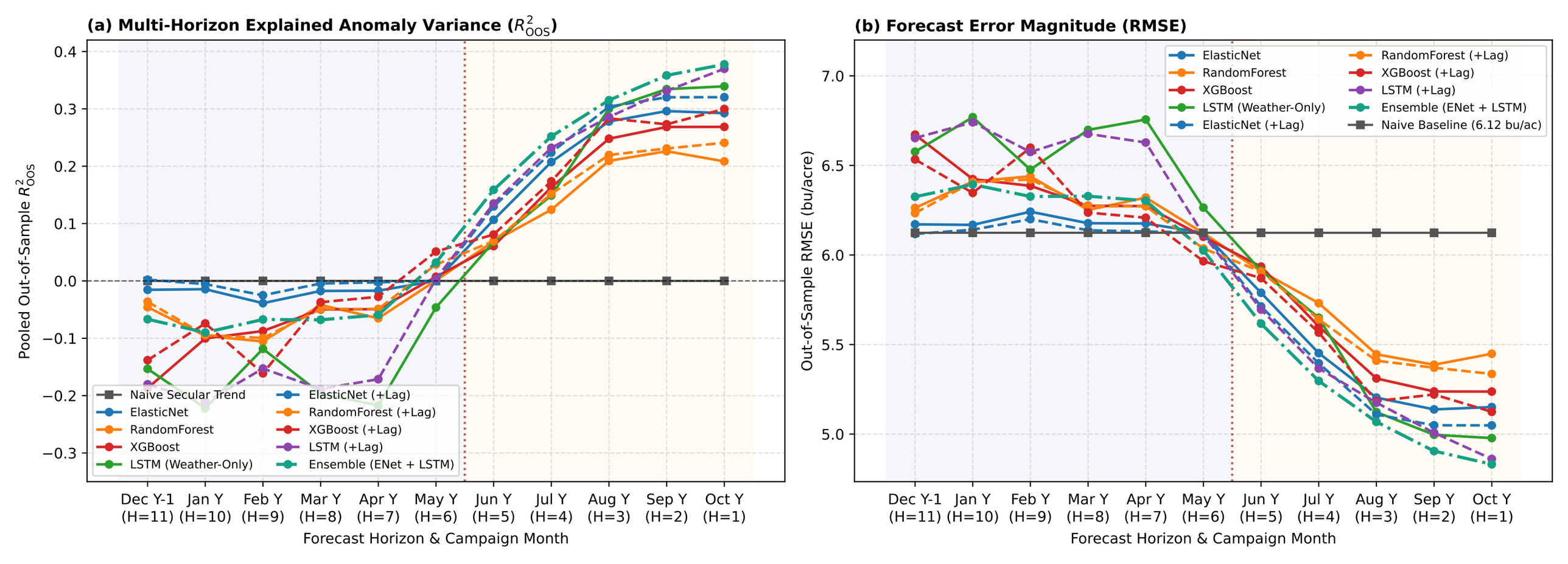
(a) Multi-Horizon Explained Anomaly Variance (R^2_{OOS})
Pooled Out-of-Sample R^2_{OOS}
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Naive Secular Trend
ElasticNet
RandomForest
XGBoost
LSTM (Weather-Only)
ElasticNet (+Lag)
RandomForest (+Lag)
XGBoost (+Lag)
LSTM (+Lag)
Ensemble (ENet + LSTM)
Dec Y-1 (H=11)
Jan Y (H=10)
Feb Y (H=9)
Mar Y (H=8)
Apr Y (H=7)
May Y (H=6)
Jun Y (H=5)
Jul Y (H=4)
Aug Y (H=3)
Sep Y (H=2)
Oct Y (H=1)
Forecast Horizon & Campaign Month
(b) Forecast Error Magnitude (RMSE)
Out-of-Sample RMSE (bu/acre)
7.0
6.5
6.0
5.5
5.0
ElasticNet
RandomForest
XGBoost
LSTM (Weather-Only)
ElasticNet (+Lag)
RandomForest (+Lag)
XGBoost (+Lag)
LSTM (+Lag)
Ensemble (ENet + LSTM)
Naive Baseline (6.12 bu/ac)
Dec Y-1 (H=11)
Jan Y (H=10)
Feb Y (H=9)
Mar Y (H=8)
Apr Y (H=7)
May Y (H=6)
Jun Y (H=5)
Jul Y (H=4)
Aug Y (H=3)
Sep Y (H=2)
Oct Y (H=1)
Forecast Horizon & Campaign Month